Casey Barton

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PROFILE

Applied AI engineer in Honolulu. I build, tune, and evaluate AI products that have to earn trust in production: define what good looks like first, put guardrails and human-in-the-loop control around the risky parts, test the failure modes on purpose, and ship only what the eval harness signs off on.

Three years running an evidence-grade agent platform: LLM evaluation harnesses and verification gates, safety guardrails, fail-closed recovery, A/B-tested prompt and configuration tuning validated before anything ships, performance monitoring, staff-level platform ownership across integrations and data surfaces. Remote-first.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · webhooks · Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · containers

AI Eval & Reliability: LLM evaluation harnesses · offline/online experiment design · A/B testing · safety guardrails · human-in-the-loop control · failure-mode & reliability testing · verification gates · audit receipts · fail-closed recovery

AI Build: LLM agent frameworks · function calling · RAG · prompt & context engineering · MCP protocol · LLM tuning via prompt/config optimization · performance monitoring

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Build, tune, and evaluate AI products across a **1,395-repository** platform: evaluation harnesses and verification gates gate every ship — nothing reaches production without an offline eval pass and a green verification check.
- Safety guardrails and human-in-the-loop control on every consequential agent action: typed state transitions, approval gates, audit receipts, recoverable runs — the guardrail pattern this role's safety questions describe.
- Failure-mode and reliability testing as standing practice: failure-injection recovery drills, idempotent resumable runs, per-failure root-cause quarantine — **fail-closed** behavior verified, not assumed.
- Tune LLM agents via prompt and configuration optimization validated by eval before ship: A/B-tested tuning cycles against held-out checks, configuration deltas tracked with experiment records.
- Performance monitoring with A/B testing across a **989-test** control plane — **62/62** verification checks green; ship/no-ship decisions made from measured outcomes, not instinct.
- Integrate **65 MCP connectors** plus APIs, webhooks, and data pipelines across customer and product surfaces — the integration surface that feeds eval, monitoring, and product analytics.

EDUCATION

University of Hawaii at Manoa 2016

Bachelor of Science

AWS Cloud Institute 2025

Cloud Computing program

Colorado State University

Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass.

GlacierEQ estate — agent systems portfolio

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Connects multiple technical surfaces — agents, APIs, databases — into one coherent, verifiable working system.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI Eval & Reliability: LLM evaluation harnesses · A/B testing · safety guardrails · human-in-the-loop control · failure-mode & reliability testing · verification gates · fail-closed recovery

AI Build: LLM agent frameworks · function calling · RAG · MCP protocol · prompt & context engineering · LLM tuning via prompt/config optimization

Data & Infra: Supabase · Postgres · SQL · REST APIs · webhooks · Vercel · GitHub Actions · containers

Practice: experiment design · performance monitoring · audit receipts · idempotent automation

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.